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Incorporating Bicycle Activity And Vehicle Travel Reduction From Bicycle Infrastructure Into Strategic Planning Tools

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INCORPORATING BICYCLE ACTIVITY AND VEHICLE TRAVEL REDUCTION FROM BICYCLE INFRASTRUCTURE INTO STRATEGIC PLANNING TOOLS

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16. Abstract

Planners and modelers at all levels of government have been grappling with the problem of how to better represent bicycling when modeling and forecasting future scenarios. Planners want to know how and to what extent infrastructure projects can increase bicycling activity and reduce the health, environmental, and financial burdens imposed by single-occupant vehicle travel. This technical transfer report describes the development of an extension to existing tools that allows strategic assessment of changes in total vehicle miles traveled (VMT) due to changes in bicycle network accessibility. The base tool corresponds to the household-level models used in the Multimodal Module of VisionEval (VE-MM), a strategic planning model developed by the Oregon Department of Transportation (ODOT) and the Federal Highway Association (FHWA). This initial development relied on readily available travel data from the Oregon Household Activity Survey (OHAS), but the methodology is designed to accept estimation data from other sources.

The final tool developed here—a collection of models providing a VE-MM compatible link between bicycle network accessibility measures and household VMT—will be useful as a proof of concept and for further justification and development of VE-MM bicycle scenario planning and policy support capabilities. We found significant, meaningful sensitivity of VMT to increases in household bicycle network accessibility. For a 1% increase in various measures of bicycle network accessibility, models underlying the tool predict a corresponding 0.025% to 0.23% reduction in daily household motorized VMT. Even greater reductions were predicted among bicycle-owning households (used as a proxy for willingness to bike). Suggestions for applying the tool at different levels of complexity and capability are provided, as well as an online technical appendix. Future directions for further development of the tool and potential integration into VE-MM are outlined.

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DISCLAIMER

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EXECUTIVE SUMMARY

This technical transfer report describes the development of an extension to existing tools that allows strategic assessment of changes in total vehicle miles traveled (VMT) due to changes in bicycle network accessibility. The base tool corresponds to the household-level models used in the Multimodal Module of VisionEval (VE-MM), a strategic planning model developed by the Oregon Department of Transportation (ODOT) and the Federal Highway Association (FHWA) (Wang, 2018; Wang et al., 2018). This initial development uses readily available travel data from the Oregon Household Activity Survey (OHAS), but the methodology is designed to accept estimation data from other sources.

Motor vehicle use is associated with a range of well-known negative externalities affecting human well-being and sustainability. Subsequently, transportation agencies across the country have focused on finding strategies to divert daily vehicular trips to non-motorized sustainable transportation options. Studies have suggested positive associations between bicycle use and bicycle network accessibility, where accessibility is measured as some combination of the quality and distance of connections to common destinations (Broach & Dill, 2016, 2017; Schoner & Levinson, 2014). However, the current version of VE-MM does not include a direct link between bicycle diversion and bicycle network accessibility, limiting its ability to evaluate related policy options.

This document explains the development of a framework for linking bicycle network accessibility measures to bicycle use and household VMT estimates, and reports initial results from the resulting models for Oregon. The new models build on existing specifications of VE-MM, but incorporate bicycle accessibility measures both directly and indirectly (via bicycle trips) as independent variables. Additionally, the proposed approach further extends VE-MM by measuring accessibility within defined distances around a household's home zone, rather than only within the zone itself.

We conceptualized the link between bicycle networks and motor vehicle travel diversion as follows. A household's home zone (e.g., block group) has a certain generalized measure of bicycle access based on the quality of connectivity to nearby destinations. Network access then combines with individual household demographics and other measures of the home zone's built environment (e.g., density, land-use mix, transit service, etc.). Controlling for the other factors, increases in bike network accessibility will tend to reduce household driving.

The tool developed here is a collection of models that provide a way to link changes in bike network accessibility to household VMT, either directly or indirectly via changes in the number of household bike trips (which then depend on bike network accessibility, among other factors). We attempted to keep the tool consistent with existing versions of

VE-MM in terms of methodology and data inputs and outputs. The developed models provide a proof of concept that can be used for further development, model estimation, and incorporation into the VE-MM software.

Existing estimation of the VE-MM tool has relied on National Household Travel Survey (NHTS), Smart Location Database (SLD), and additional sources for network supply measures (Wang, 2018; Wang et al., 2018). To operationalize the conceptual model for initial development of the tool, we used the following data sources:

- Household VMT and demographics: OHAS 2010-11
- Home zone built environment: SLD 2014
- Bike network accessibility:
    - People for Bikes Bicycle Network Analysis (https://bna.peopleforbikes.org/,accessed November 2021)
    - University of Minnesota Access Across America: Biking 2017 (https://access.umn.edu/research/america/biking/2017/)
        - All bikeable facilities
        - Level of Traffic Stress (LTS, Portland region only)
    - Portland State University Route Quality Index 2017 (Broach & Dill, 2017)

We developed models that extended the existing VE-MM household VMT model to include various bike network access measures. In addition, we estimated models of bike trips—which have stronger research support linking to network measures—that could then inform household VMT models. Finally, we selected our preferred initial models to create a usable tool that could estimate household VMT responses to various bicycle network improvement scenarios. In testing reasonable hypothetical planning scenarios, we found meaningful sensitivity of VMT to increases in household bicycle network accessibility. For a 1% increase in various measures of bicycle network accessibility, models underlying the tool predict a corresponding 0.025% to 0.23% reduction in daily household motorized VMT. Even greater reductions were predicted among bicycle-owning households (used as a proxy for willingness to bike).

The final tool developed here—a collection of models providing a VE-MM compatible link between bicycle network accessibility measures and household VMT—will be useful as a proof of concept and for further justification and development of VE-MM bicycle scenario planning and policy support capabilities. Future development could include:

- estimation on a national dataset (including potential expansion of the PSU and UMN datasets);
- coding development of a bicycle network VE-MM module built on this tool;

- further consideration of household types and propensities to bicycle;
- development of a graphical user interface to better present and understand the bicycle network accessibility policy "levers" and outcomes; and
- further exploration of more flexible model forms for household VMT estimation, including potential joint estimation of bicycle trips and household VMT.

The remainder of this report provides an expanded description of the core tool that was developed. Additional technical details and example code are provided as a digital appendix, and a link is provided at the end of this document.

1.0 BACKGROUND

Motor vehicle use is associated with a range of well-known negative externalities affecting human well-being and sustainability. Subsequently, transportation agencies across the country have focused on finding strategies in which daily vehicular trips can be diverted into non-motorized sustainable transportation options. Until recently, non-motorized travel was incorporated in transportation planning models only in a limited way, if at all. Over the past decade, this trend has begun to change rapidly. Along with well-documented relationships to land use, urban form and demographics, bicycle networks are increasingly recognized by research as crucial components in explaining bicycling activity. Network improvements are the focus of many policies and plans that seek to increase biking (Buehler & Dill, 2016). Therefore, it is necessary for current transportation planning to incorporate bicycle networks into consideration.

While a handful of regional travel models around the U.S. have incorporated bicycle network accessibility, strategic planning models provide perhaps the most compelling case for better integrating potential shifts to bicycling. These models are typically focused on a specific planning outcome or set of outcomes, and use a simplified subset of input variables that are easier to assemble and forecast. However, to date, strategic models have focused on motorized traffic and typically incorporate walking and biking only as fixed inputs. Among them, the VisionEval (VE) family of models has been developed by a broad consortium of supporters, including U.S. DOT/FHWA, AASHTO, and ODOT. VE is currently used to assess light-duty transportation-related emissions and was originally known as Greenhouse Gas Strategic Transportation Energy Planning (GreenSTEP). VE grew out of the need to coordinate various spin-offs of the GreenSTEP model system, including the Regional Strategic Planning Model (RSPM), Rapid Policy Assessment Tool (RPAT), and Emissions Reduction Policy Analysis Tool (EERPAT), and to make the development of the model system easier or more accessible. However, VE does not escape the trend of limited non-motorized capabilities and incorporates changes in biking and walking mainly as fixed inputs and with no connection to policy changes that improve non-motorized network accessibility.

Efforts to expand the reach of these models were put into the conception and development of the Multimodal Module (VE-MM). The aim of this module was to understand how traveler behavior could change due to policy and investment decisions and that could characterize the intermodal shift. The module uses behavioral models incorporating the 2009 National Household Travel Survey (NHTS) for travel and sociodemographic data, EPA's Smart Location Database (SLD) for built environment attributes, and regional roadway and transit services for transportation supply measures.

The tool developed in this project extends VE-MM to estimate household VMT responses to improvements in bicycling networks and accessibility. The tool leans heavily on methods from emerging research on bicycle network definitions and links between network quality-weighted access and diversion of trips to bicycling.

2.0 TOOL DEVELOPMENT

2.1 CONCEPT

VE-MM models household VMT as a function of household demographics, the built environment of the household's home zone (e.g., block group), and the regional supply of transit service and freeways. **Error! Reference source not found.** describes how we conceptualized the link between bicycle networks and motor vehicle travel diversion within the existing VE-MM framework. A household's home zone (e.g., block group) is assigned a generalized measure of bicycle access based on the quality of connectivity to nearby destinations. Network access is then combined with individual household demographics and other measures of the home zone's built environment (e.g., density, land-use mix, transit service, etc.). Models are then developed to predict the impact of bike network accessibility on household driving, either via a direct reduction in VMT or an indirect reduction via an increase in household bicycle trips. Our initial tool omits regional system supply measures (e.g., freeway miles and transit service) due to the limited number of regions within our Oregon dataset that had available data.

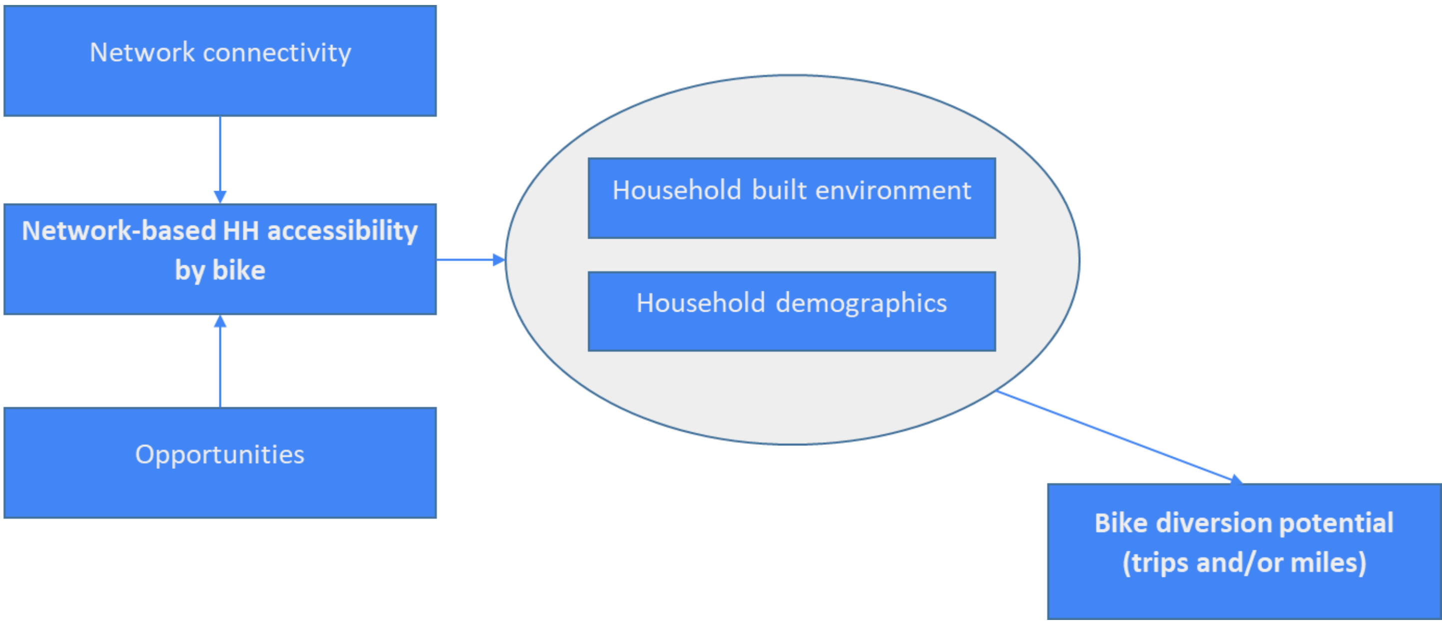

Figure 1 Operational conceptual model

The tool developed here is a collection of models that provide a way to link changes in bike network accessibility to household VMT, either directly or indirectly via changes in the number of household bike trips (which then depend on bike network accessibility, among other factors). We attempted to keep the tool consistent with existing versions of VE-MM in terms of methodology and data inputs and outputs. The developed models provide a proof of concept that can be used for further development, model estimation, and incorporation into the VE-MM software.

2.2 DATA

To operationalize our concept into a tool we explored numerous data sources, including those used by the existing VE-MM module, and eventually selected those shown in Table 2.1. Initially, we were uncertain about the national bike network measures (BNA and UMN-A) and decided to limit our initial analysis to Oregon, where we could fall back on the PSU network measure that had existing research supporting links to behavior. OHAS, the statewide travel survey, provided a large enough sample of Oregon travel to estimate our models. Similar to VE-MM, we relied on SLD for standardized built environment measures at the block group scale.

Table 2.1 Data sources

Variable group	Source(s)	Description
Household VMT	OHAS 2010-11	Total VMT on survey day
Household demographics	OHAS 2010-11	Post-processed to match existing VE-MM variables
Built environment	SLD 2014	Block group level
Bicycle network accessibility	BNA (Nov 2021)	Block group index (0-100)
	UMN-A 2017	All bikeable facilities, retail & all jobs
	PSU (2017)	Route Quality Index (RQI)

2.2.1 Household data

While VE-MM has traditionally relied on the National Household Travel Survey (NHTS), we developed our concept using the Oregon Household Activity Survey (OHAS, 2011) both for convenience and sample size considerations. OHAS does not collect data on annual household VMT, so we used VMT on the survey day as a proxy. A few other variables had to be constructed to match NHTS as closely as possible:

- Lifecycle: parent with/without children, empty nester, and single
- Number of driving licenses
- Cars per license
- Number of workers
- Income
- Number of people by age group (<16, 16-65, >65)

Household size was not included because it was highly correlated with other variables.

2.2.2 Land-use and built environment data

The SLD is built environment data collection for all of the United States constructed by the EPA. The measures are constructed at the block group level. These measures are from the same sources as those used in VE-MM:

- D1b: Gross population density [ppl/acre]
- D2a_WrkEmpHousehold: workers per household
- Dbpo3: Intersection density
- D4c: Aggregated frequency

Since the household sample is entirely in Oregon, Census regions are not included as predictors in this work.

2.2.3 Bicycle network accessibility measures

With regard to bicycle accessibility measures, there is no consensus on the appropriate method of measuring. We identified three existing measures to test:

- People for Bikes Bicycle Network Analysis (BNA) index
  - Available nationally in most urban areas
  - Summarizes access on a low-stress network to a range of common destinations at the Census block level

- University of Minnesota Accessibility Observatory Lab's Access Across America Bike access measure (hereafter, UMN-A)
  - Block level
  - Full measure publicly available for the 50 most populous U.S. regions; potentially available anywhere in the U.S.; the simplified version available nationwide
  - Summarizes bike access to jobs in different categories over a range of travel times (and, in the full version, traffic stress levels) at the Census block level

- Portland State University Route Quality Index (PSU-RQI, Broach & Dill, 2017)
  - Based on published research, but only currently calculated for the Portland metro region
  - Summarizes expected quality of access to jobs at the Census block group level; designed to extend to any type or types of destination

All three measures are sensitive to changes in bicycle network connectivity and quality. Each method works by solving "best" (lowest stress or highest utility) routes from origin zones to a fixed set of destinations representing opportunities. BNA and UMN-A consider only portions of the total network based on variations of the Level of Traffic Stress (Mekuria, 2012) and then solve the shortest paths. The PSU-RQI measure takes a different approach, calculating the least-cost routes between points using the entire network and then calculating relative quality along the representative route (Broach & Dill, 2016). BNA and UMN-A use OpenStreetMap (OSM) networks, while PSU-RQI relies on a Portland metro -style bike network. BNA uses a "market basket" of common destinations, awarding points when one or more examples of a destination type (e.g., parks or health care facilities). UMN-A and PSU-RQI use Census jobs data.

Table 2.2 summarizes the measures as applied to our Oregon statewide sample.

Table 2.2 Summary of bicycle network accessibility measures used

Measure*	Mean	Min	Max
BNA (overall score, 0-100)	27.8	0.00	96.8
UMN-A LTS4, 20-min, all jobs (1000s of jobs)	38.4	0.00	259.2
Portland region only			
UMN-A LTS2, 20-min, retail jobs (1000s of jobs)	2.74	0.00	17.0
PSU-RQI (0-1.2, where 1.0 equiv. to striped bike lane along entire route to average job w/in 5 miles)	0.69	0.23	0.86

* all measures were summarized to the block group level for comparison

2.3 MODELING

Bicycle network accessibility was added to the adapted VE-MM household VMT model using two types of specification. First, directly as bicycle accessibility. Second, through separate estimation of bike trips.

1. Daily VMT = f(household demographics, household built environment, bicycle network accessibility)

2. Daily VMT = f(household demographics, household built environment, bicycle trips=f(bicycle network accessibility))

As the scope of the work here was restricted to a proof of concept with data from one state, some adjustments were made to the specification. Specifically, some variables were omitted, as previously explained in the data section.

2.3.1 Base model

Table 2.3 shows the differences between variables used in VE-MM and the bicycle extension tool base models.

Table 2.3 Comparison of base model specifications (VE-MM vs. Tool)

Variable	VE-MM	Bicycle extension
Household		
Three age categories [<16,16-65,>65]	✓	✓
Household driver licenses	✓	✓
Number of workers	✓	✓
Income (log)	✓	✓
Vehicle per driver (log)	✓	✓
Lifecycle	✓	✓
Household size	✓	
Built environment		
SLD: D1B (pop. density)	✓	✓
SLD: D2A_WRKEMP (jobs/housing balance)	✓	✓

SLD: D3bpo (intersection density)	✓	✓
SLD: D4C (transit service frequency)	✓*	✓
Census region	✓	
FWYLaneMiPC	✓	
TranREvMiPC	✓	

* only as interaction term with TranREvMiPC

Coefficient direction and significance lined up well between the two base models, increasing our confidence that the Oregon/OHAS-based data were broadly compatible with the VE-MM estimation data, at least conceptually. Model performance comparisons should be taken with a grain of salt for several reasons:

- VE-MM used the sum of approximated average annual vehicle miles (AADVMT) traveled from NHTS, while our estimation relied on VMT on a single travel survey day.
- VE-MM used a specific power transformation of VMT (VMT^0.38) optimized for the dataset; our estimation duplicated that specification but also estimated a specification tuned to our dataset.
- As noted, there were differences in the predictor variables.

VE-MM reported an R^2 of 0.47 and RMSE of 29.3 for the urbanized area models. Our base model estimation (without bike accessibility measures and for the VE-MM power transform and a log-linear model, respectively) resulted in R^2 of 0.27-0.29 and RMSE of 35.1-36.2, broadly in line with the moderate level of fit of the VE-MM nationwide estimation. The remainder of VMT results discussed are from log-linear models that used the original household survey weights.

2.3.2 Direct VMT models with bike network accessibility

Only the BNA score and UMN-A LTS4 (all bikeable streets) were available for the statewide sample. The overall BNA score (0-100) for the area around the household was significant ($p<0.05$), with the expected negative impact on household daily VMT. We also tested BNA alongside the more general UMN-A 20-minute job access score. In combination, model fit improved slightly and the two were jointly significant; however, BNA score was then individually significant only at a higher threshold ($p<0.10$), and the marginal effect of BNA score was diminished.

We also tested models using only the subgroup of households that reported owning a bicycle, with the idea that the impact of bicycle networks is likely greater for households that have shown some basic willingness to bicycle. Intuitively, bike-owning households (just over half the OHAS sample) demonstrated a stronger VMT response to improved BNA scores. Current versions of VE-MM do not include household bicycle ownership or other measures of bicycling propensity, but we provide the results here to inform future development efforts.

Table 2.4 Estimated marginal effects of changes in BNA score: direct daily VMT models

Model specification	Expected VMT impact of one-unit BNA increase (mi/day)	Expected daily VMT impact of 1% BNA increase
Base + BNA	-0.021	-0.034%
Base + BNA + UMN-A	-0.015	-0.025%
Base + BNA + UMN-A: Bike-owning households	-0.036	-0.115%

Table 2.4 summarizes estimated marginal effects that will be reflected in the tool. While the marginal impacts on VMT initially may appear small, they suggest that substantial shifts in BNA score could lead to meaningful reductions in daily VMT. For example, improving BNA by 25 points (about the difference between average conditions in Boise, ID, and more bikeable Eugene, OR, as of this writing) would lead to an expected decrease of a half mile per day per household, or nearly one mile per day in bike-owning households. This is an average effect estimate, and another way to interpret it would be that every fourth household replaced one daily two-mile vehicle trip with biking, which strikes us as a plausible level of diversion for considerable bike network improvement.

2.3.3 Indirect VMT models via changes in bike trips

We also tested models of bike trips per household as a function of BNA and UMN-A network accessibility scores. Daily VMT was then modeled sequentially with predicted bike trips as an input. Existing research on bicycle network access has largely focused on trips and not miles traveled.

When included in the VMT model, a household's number of bike trips was a significant (negative) factor in household VMT. We then developed separate count models of daily bike trips as a function of BNA and UMN-A (LTS4, 20-min, all jobs), and fed the predicted values into the daily VMT models. BNA and UMN-A were significant (positive) factors in daily bike trip count, and predicted bike trip count was a significant (negative) factor in household VMT estimation. We also considered this chain of impact separately for bicycle-owning households. Calculated effects are shown in Table 2.5.

Table 2.5 Estimated marginal effects of changes in BNA score: indirect daily VMT models via predicted bike trips

Model specification	Expected VMT impact of one-unit BNA increase (mi/day)	Expected daily VMT impact of 1% BNA increase
Base + BNA	-0.032	-0.090%
Base + BNA + UMN-A	-0.027	-0.082%
Base + BNA + UMN-A: Bike-owning households	-0.310	-2.547%

Estimated impacts of changes in BNA on VMT were noticeably larger when estimated via bike trips than in the direct VMT models. In particular, bike-owning households had an elastic (greater than 1:1) response to better bike network accessibility, with an expected VMT decrease of about 2.5% for each percent increase in BNA. While we emphasize the several caveats around this preliminary analysis—sequential model form, relatively poor explanatory power of bike trip models, challenges in identifying bike-owning households—the results suggest understanding diversion via trips instead of direct VMT diversion is worth exploring.

2.3.4 Portland region models

For a sample of around 3,700 households in the Portland region, we had access to additional bike access data from both UMN (multiple stress levels) and PSU (route quality index, RQI). We tested numerous iterations of the UMN-A scores by stress level (LTS 1-3), job type (retail, all), and network bike travel time (5-30 minutes). We used the PSU measure as it was presented in the original research, as a multiplier on jobs within five network miles.

In the Portland subsample, BNA and PSU scores were not significant in the direct VMT diversion model. The UMN-A access score was significant, with the best-fitting model using job access using their LTS2, 15-minute, retail jobs metric. LTS2 corresponds to lower-speed (<= 30 mi/hr) streets with striped bike lanes or residential streets (<= 2 lanes, and <= 25 mi/hr) in the UMN-A typology (https://cts-d8resmod-prd.oit.umn.edu/pdf/cts-19-20.pdf). All three access measures were significant predictors of daily household bike trips. Estimated effects are provided in Table 2.6.

Table 2.6 Estimated marginal effects of changes in bike access scores: Portland-only models

Model specification	Expected VMT impact of one-unit access score increase (mi/day)	Expected daily VMT impact of 1% access score increase
Direct VMT		
UMN-A (LTS2, 15-min, retail jobs/1000)	-0.740	-0.068%
Indirect VMT via Bike Trips		
BNA	-0.029	-0.088%
UMN-A (LTS2, 20-min, retail jobs/1000)	-0.099	-0.035%
PSU-RQI (/100)	-0.057	-0.226%

UMN-A showed the most promise for capturing direct VMT diversion. In the indirect models, the PSU measure showed the greatest estimated leverage (-0.2% VMT per %PSU-RQI score), followed by BNA and then UMN-A. The indirect PSU model of VMT was also the best-fitting, although the differences were not large. Given the mixed results, we would recommend consideration of all three access measures in future work. The PSU-style bicycle network needed to calculate the access score is currently available only for Portland, Eugene, and Bend in Oregon, and that would pose a challenge especially for nationwide expansion. The expanded UMN-A scores are publicly available only for the top 50 metro areas across the U.S., but based on conversations with the maintainers, the scores could be calculated for any area in the U.S.

3.0 USING THE TOOL

The tool presented here consists of a statistical methodology that translates specific measures of bicycle network accessibility into expected changes in household daily VMT. The bike access measures tested account for both network quality and proximity to opportunities. We constructed the tool to enable eventual integration into the VisionEval Multimodal (VE-MM) strategic modeling system, but for now it stands alone. Users will likely apply the current tool in one of three ways:

1. At the simplest, as a sketch planning method for understanding likely VMT impacts for given bicycle network access improvements.
    - E.g., "If we increase BNA scores by 10% for our region (or state, locality, etc.), what is the likely impact on household VMT?" Considering the range of models estimated for our Oregon sample, the answer would be somewhere from a 0.25-0.9% VMT reduction.
    - These sketch results could further be extrapolated using basic local data into annual VMT reduction, and, using assumed VMT ratios, carbon or other emission reductions.
2. Scenarios can be simulated by applying a stated change in bicycle network access across the sample (or subsample; e.g., by geography or demographics) to test various policy proposals.
    - A base scenario and change would be calculated by one of the VMT models to households in a sample (the estimation sample or a local or simulated sample).
    - Changes to bike network access would be relative to a measured or assumed base level, and could be applied asymmetrically across households; (e.g., increasing each household to the next higher access score quartile, or increasing the scores in specific sub-regions).
    - Unlike method #1 above, this method would attribute VMT changes to specific households, allowing for distributional analysis.
    - This type of analysis would most readily be undertaken in R or a similar statistical software package in the current tool version.
    - We provide an example (shown below) with code in the online technical appendix that compares the current scenario to the hypothetical case where sampled Oregon households had no low-stress bike access to key destinations.

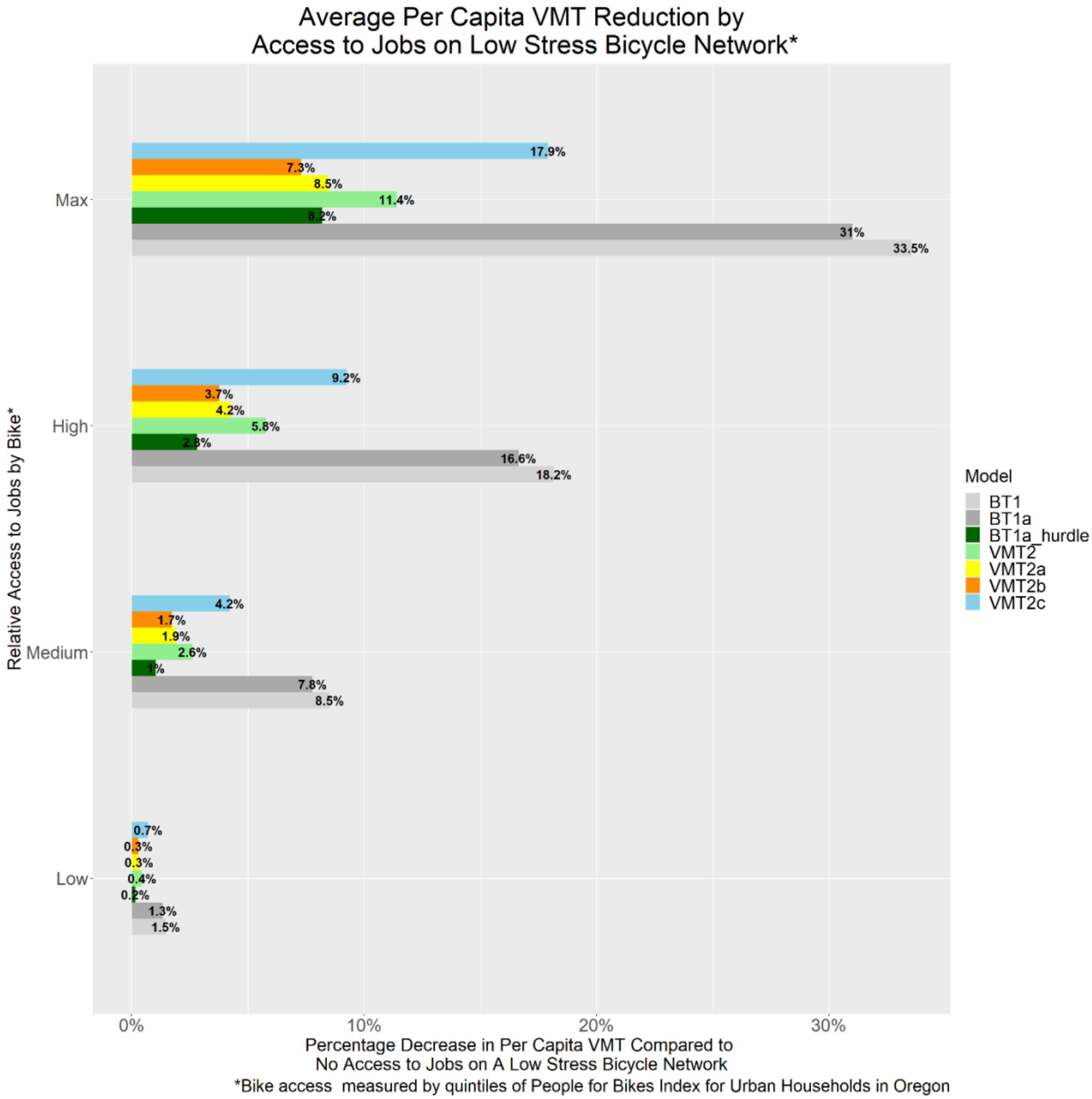

Figure 2 Simulated impact of existing low-stress bicycle networks in our Oregon statewide sample (see online technical appendix for details and descriptions of the various VMT (direct) and BT (indirect) models applied.)

3. Users with more advanced network data and analysis capability could generate modified networks and calculate access scores for different scenarios, rather than simply asserting relative changes in bike network accessibility.
   - All of the access scores tested are open and fully documented.
   - UMN-A requires only an OSM network and Census employment data.
   - BNA requires a modified OSM network and a place directory.

    - PSU-RQI requires a specific network format including: bike facility type, intersection controls, roadway slope, and estimated traffic volumes, along with Census employment data.

4.0 LIMITATIONS, CONCLUSION AND DISCUSSION

This report and associated online technical appendix describe the initial development of a tool that links strategic bicycle network and infrastructure policy to motorized VMT reductions. The tool is designed to be further developed and integrated into the strategic planning framework VisionEval, specifically within the multimodal module, where it could improve on current bicycle capabilities. As a standalone tool, the methodology and estimates developed in this project serve as a usable proof of concept for various bicycle network and VMT reduction policy applications, and we hope it provides the groundwork for future refinements, expansions, and extensions.

As a limited proof of concept, the tool has several known limitations that should be considered before use:

- The estimation sample was limited:
    - covered only households within the state of Oregon
    - dated from 2010-11
    - captured only a single day of travel;
- Underlying models' explanatory power was only moderate ($R^2 < 0.40$);
- Indirect VMT via Bike Trips models were estimated sequentially, with no statistical representation of likely overlap in error between each model stage;
- None of the bike access scores tested are currently available everywhere "off the shelf," and would require additional effort to calculate where they do not yet exist;
- Our efforts to minimize changes to the existing VE-MM VMT models might have led to some additional error in variable calculation and increased the potential for mis-specification of the models. We did take some steps to mitigate those concerns where possible, but future efforts might consider additional variable and model specifications; and
- We lacked the resources to test measures of bicycle network supply separately (e.g., miles of bike lane or other facility near a household). These simpler measures are worth exploring, in addition to the more complex measures of network accessibility scores.

Despite these considerations, the current tool version demonstrates strong potential to link policy goals of reducing VMT with measurable strategies to improve bicycle networks. Based on this initial effort, we recommend:

- continuing to explore both direct and indirect (via bicycle trips) household VMT modeling pathways;
- continuing to consider multiple methods of bicycle network scoring, including those tested here (PeopleForBikes Bicycle Network Analysis [BNA], University of Minnesota Access Across America: Bike Access, and Portland State University's Route Quality Index [RQI]);
- exploring available applications of the current tool, to better assess strengths, weaknesses, and usefulness in different contexts; and
- identifying opportunities to further develop the tool, including:
  - re-estimation on a national sample (e.g., NHTS)
  - modeling refinements
  - tool extensions that increase usability, such as a graphical interface for scenario testing
  - testing and integration with VE-MM.

5.0 REFERENCES

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6.0 APPENDICES

APPENDIX A

ONLINE TECHNICAL APPENDIX

An online technical appendix is provided as a GitHub repository and website. The pages there provide the code underlying tool development and full model specifications that constitute the tool itself. Examples are also provided for more advanced application. Information in the Technical Appendix will be updated with any corrections, clarifications, and future development. The appendix may be accessed at the following link:

https://github.com/jbroachpdx/bike-network-vmt-tool